

Let's go back on line powerflow equations. They allow us to compute the power that goes from one node i to another j .

Remember that current and voltages values are sinusoidal, so the power has this shape as well. The active power is the actual power consumed by the loads, while reactive power is the power oscillating.

Let's focus on the active power P_{ij} .

$$P_{ij} =$$

G_{ij} and B_{ij} are line parameters. The power depends on the length of the cable, the size, the material, etc. This is considered in the parameters.

V_i is the amplitude of the voltage at bus i . $\theta_{ij} = \theta_j - \theta_i$. θ is the phase of the voltage signal. By plotting voltage on a given bus i , we'll have a sinusoidal voltage with a given amplitude. Doing a similar thing for a bus j , we'll also have a sinusoidal form, but it will be shifted from the one of bus i , and θ stands just for this.

The oscillations frequency should be bounded to 50 Hz.

This is also a mathematical model of a physical system.

It's different from the swing equations, as in that we have two differential equations, which we will see as state space equations/model.

The one for power have no derivatives, and are algebraic equations. This will model a static system, while swing equations describe a dynamical system.

So, so far we've seen the generators and the power lines.

Remember that from the transportation network we go to the distribution network. Most of the cables in Rome are distribution network cables.

The connection from transmission networks to distribution network takes place with a primary substation, reducing the voltage level from high voltage to medium voltage.

Inside the distribution network we have another type of distribution network given by the low voltage distribution network, working at 100V for example.

The boundaries between these networks are critical locations, taking the name of substations. As we've seen, the boundary between the high voltage and medium voltage is called primary substation, while between the medium voltage and low voltage is called secondary substation. We can also identify the boundary between generation and transmission network, taking the name of Generation substation.

Medium voltage cables take the name of "feeders". Feeders bring power to lower voltage distribution network, having once another transformer reducing the amount of power transmitted then to premises. On poles, we can also find a switch/breaker, that we can use to switch on and off cables. It could be a potential target of attacks as well.

Let's go back to the sensors used for the transformers in primary substations. We have three categories of important devices available in primary/secondary substation:

- Sensors: like the PMU (Phase Measurement Units), allows us to measure the phase shift of signals. We also have current and voltage sensors.
- Actuators: Like breakers/switches, used to connect and disconnect devices in the substation. The transformer itself is an actuator as well.
- Logical controllers (micro controllers): Taking the name of IED (Intelligent Electronic Devices). They take data from sensors, and take decisions on actuators.

What is a typical layout of medium voltage network? The big black box is the connection with the high voltage grid. Everything up to L_1 is a primary substation (also the lower part). We have two transformers (where the double circle concatenated is a symbol for the transformers). They transform power from 230 kV to 22 kV.

In the station we also have a circuit breaker. After the transformer, before the start of the medium voltage line, we have the breaker. We must be able to disconnect cables from transformers in case it's necessary. The circuit breaker is commanded by the IED.

Medium voltage cables bring electricity to customer, loads (L), and the symbol is given by an arrow.

We also have remote terminal units, which are controllers equipped with sensors and sometimes actuation functionalities.

We have then once another interruptor, with the label "NOP" which means that the switch is normally open. Thus, normally the medium voltage grid is "radial", having no loops. These switches are important in the case of faults, creating a new path for customers which would be isolated. This procedure takes the name of FIST Fault Isolation and Service Restoration procedures.

Let's take a look at IEDs. It has inside two main functionalities:

- sensors, able to measure the current I , the voltage V , the frequency f , the active power P , the reactive power Q .
- internal protection logics: telling the electronical devices what to do depending on measurements.

They are responsible for commanding switches and breakers.

In the traditional power grids, the generation is centralized and fully controllable. Power flows unidirectionally. Medium voltage and low voltage are passive, having no generation there.

Nowadays, even in medium voltage grid and low voltage grid we have distributed generation, with solar power, wind turbines, etc. Contrary to centralized generation, this is not totally controllable. We talk about smart grids.

We have massive introduction of smart loads as well, often connected to the internet.

Once another difference is given by the ability of storing electricity through batteries.

One important algorithm is given by forecasting, determining how much power we will generate from renewable sources in the coming days, this is necessary to keep balance.

Nowadays, some buildings, are already equipped with smart loads, thus operators are able to control loads, determining how much energy it has to consume in a second. These algorithms are connected to the internet, so somebody could take control of these algorithms, performing a Load Altering Attack. These algorithms are known as BMS, Building Management Systems.

Operators could use these algorithms to save money (reducing the amount of power consumed without impacting users), and also to increase user comfort.

To recap the main actors, we have:

- GENCO
- TSO
- DSO
- Retailers: provide electricity to premises.
- Aggregators: To sell electricity in the energy market, we need to have huge generators. We talk about bulk generators. The idea of virtual power plants is to put together a lot of small generators (of the people, even if they're far away

geographically) acting like they're one big generator. This will allow to sell energy.

- Market operators: companies responsible for operating the energy market. Similar for transmission and distribution companies, but unlike them they're responsible for the fair execution of the market.
- Customers.

We have several types of markets. We can distinguish energy markets and flexibility markets. They're algorithm systems in which we buy and sell electricity (electricity markets).

Dive in flexibility services, I got a bit lost.

In flexibility markets system operators buy energy to keep the power balance. Energy for balancing is more expensive than the energy we consume.

These are two groups of markets. We have several energy markets and flexibility markets. They're automatic algorithms and are also targets of attacks. By attacking the market, we want to affect the cost of energy, for example.

Now, let's focus on control rooms.

Every company has a role, performed by an algorithm, and they have control centers in which these algorithms run on a computer. Sensors will send data to the control room, which will give commands sent to actuators.

What is the layout of a control room? In the control room, we have the SCADA system, which means Supervisory Control And Data Acquisition. It's the set of all infrastructures used to take data from sensors, run algorithms (supervision) and take control decisions (control).

In control rooms we have big screens including a schematic representation of the network (like for TSO we have the scheme of transportation network), with values. We also have workstations hosting different algorithms, like state estimation algorithms (taking data from sensors, estimating the current status, displaying it then on the screen).

What is the architecture of a SCADA system? It's a layered architecture.

Specifically, layer 3 and 4 are for the SCADA control room. Inside of it, we have displays (human machine interface), servers running algorithms, like the "historian" server, which keeps track of everything with a log.

At the bottom we have the actual system that is under control, taking the name of "field". The first layer, attached to the physical system, is composed by remote terminal units (RTU), and PLC (programmable logic controller). Controllers send data to higher level controller, and through wireless/wired network it will reach

the control center.

RTUs may also perform local decisions in certain cases, without waiting for commands from control centers.

Most of the attacks take place in the middle, so in the transmission of data. While other may target the firmware of RTUs. Others the databases.

Up to L4 (only including the auth. servers) take the name of Operational System.

The upper level is called the corporate network.

Nowadays, also, corporate network and operational network are connected to the internet.

One important algorithm running in the SCADA is called the distribution management system (DMS). It's goal is to solve the powerflow equations. For example, it's used to perform simulations. Imagine that an operator has to disconnect a cable (for maintenance). Every time we need to do an action like this, we simulate it on computers first.